Duckiebot Command Cheat Sheet

SSH | Linux | scp | git | make | ROS2 Humble | tmux | Docker

Replace `duckieXX` with the number written on your robot. Replace `yournick` with your own nickname.

EMERGENCY STOP

Robot running away? Open any shell in the container and run:

```
ros2 topic pub --once /duckieXX/wheels_cmd
  duckietown_msgs/msg/WheelsCmdStamped "{vel_left:
  0.0, vel_right: 0.0}"
```

Ctrl+C stops the node you are looking at, nothing else.
space stops the robot while you are driving with the keyboard.

1 Connect to the robot

```
ssh duckie@duckieXX.local
```

Password: `quackquack`. `XX` = the number written on the robot.

2 SSH keys: stop typing the password

GENERATE — Windows PowerShell

```
ssh-keygen -t ed25519 -f .ssh/nickname -C "nickname"
```

GENERATE — mac / Linux

```
ssh-keygen -t ed25519 -f ~/.ssh/nickname -C nickname
```

COPY

```
scp .ssh/nickname.pub duckie@duckieXX.local:~/.ssh/nickname.pub
```

APPLY

```
ssh duckie@duckieXX.local "(echo; cat ~/.ssh/nickname.pub) >>
  ~/.ssh/authorized_keys && chmod 600 ~/.ssh/authorized_keys && rm
  ~/.ssh/nickname.pub"
```

CHECK — must log in with no password

```
ssh -i .ssh/nickname duckie@duckieXX.local
```

`nickname` = your own nickname; use it so students do not overwrite each other's file.
mac / Linux: `ssh-copy-id -i ~/.ssh/nickname.pub duckie@duckieXX.local` does COPY + APPLY in one command.
Only the `.pub` (public) file ever leaves your computer.

3 Linux shell basics

`pwd` — where am I
`ls` — list files here
`cd folder` — go into a folder
`cd ..` — go one folder up
`cat file` — print a file
`mkdir folder` — new folder
`touch file` — new empty file
`rm file` — delete a file
`rm -r folder` — delete a folder with everything in it
`chmod +x script.sh` — make a script runnable
`ls | grep .py` — filter the output, keep lines with `.py`
`nano file` — edit: save = **Ctrl+O**, **Enter**; exit = **Ctrl+X**
Ctrl+C = stop the running program. **Tab** = complete the name. **arrow up** = previous command.

4 Copy files — run these ON YOUR COMPUTER

```
scp myfile.py duckie@duckieXX.local:~/DuckieExamples/
```

Send one file to the robot.

```
scp -rp src/. duckie@duckieXX.local:~/DuckieExamples/src/
```

Send a whole folder. The `/.` matters: without it you get `src/src` once the folder already exists there.

```
scp 'duckie@duckieXX.local:~/DuckieExamples/images/*.jpg' .
```

Get the pictures. The quotes matter: they stop your own shell expanding `*`.

```
./get_images.sh duckieXX
```

Same thing, script in the repo.

5 git

`git config --global user.name "Your Name"` — once, before your first commit
`git config --global user.email "you@example.com"` — once, same
`git clone <url>` — download the repo
`git pull` — get the newest changes
`git status` — what did I change
`git add .` — stage all changes
`git commit -m "message"` — save a snapshot
`git push` — upload your commits
`git log --oneline` — history, one line per commit
`git diff` — show the changes line by line
`git checkout -b my-branch` — create a branch and switch to it
`git checkout my-branch` — switch to an existing branch
`git branch` — list branches
Repo: `github.com/TLF-2026-Robotic-Track/DuckieExamples`
On the robot clone with the HTTPS url, so no keys are needed there — and never `push` from the robot.

6 The project, on the robot

`export VEHICLE_NAME=duckieXX` — the robot's own name, part of every topic name
`export USER_NAME=yournick` — letters, digits and _ only: no dashes, no Cyrillic
`make build` — build the image and the workspace
`make run` — start the container and open a shell in it
`make shell` — one more shell in the same container
`make down` — stop and delete the container
`make clean` — delete `build/ install/ log/`
After editing a `.py` or a launch file you do **not** rebuild — just restart the node.
`make build` only after adding or renaming files, or changing `setup.py` / `package.xml` / requirements.

7 Run the examples (inside the container)

`ros2 run blinker blinker.py` — LEDs cycle red, green, blue
`ros2 run camera camera.py` — saves pictures into `images/`
`ros2 run robot main.py` — movements: turns commands into wheel and LED commands
`ros2 run control_room main.py` — reads the arrow keys
`ros2 launch master_launch keyboard.launch.xml` — control_room + robot together
`ros2 launch master_launch demo.launch.xml` — the same, plus the camera

Driving keys (keyboard control)

↑ / `w` forward	↓ / `s` backward
← / `a` turn left	→ / `d` turn right
`space` stop	`1` `2` `3` LEDs red / green / blue
`0` LEDs white	`q` quit and stop the robot

8 Look inside ROS2

`ros2 topic list` — all topics on the network
`ros2 topic echo /duckieXX/wheels_cmd` — watch the messages go by
`ros2 topic info <topic>` — type of the topic, who publishes, who listens
`ros2 node list` — all running nodes

Robot topics (`/duckieXX/...`)

`image/compressed` — camera
`range` — distance sensor
`tick` — wheel encoders
`wheels_cmd` — wheel commands
`led_pattern` — LEDs
`emergency_stop` — emergency stop
`temperature` — temperature

Drive with no code at all

```
ros2 topic pub --once /duckieXX/wheels_cmd
   duckietown_msgs/msg/WheelsCmdStamped "{vel_left: 0.4, vel_right:
   0.4}"
```

Wheel speeds go from `-1.0` to `1.0`; positive is forward, `0.0` stops.

All LEDs red with no code

```
ros2 topic pub --once /duckieXX/led_pattern
   duckietown_msgs/msg/LEDPattern "{rgb_vals: [{r: 1.0, a: 1.0}, {r:
   1.0, a: 1.0}, {r: 1.0, a: 1.0}, {r: 1.0, a: 1.0}, {r: 1.0, a: 1.0}]}"
```

9 tmux: keep programs running after you disconnect

`tmux new -s name` — new session
`tmux ls` — list sessions
`tmux attach -t name` — go back into a session
`tmux kill-session -t name` — delete a session

tmux: press `Ctrl+B`, release, then

`d` — detach, everything keeps running
`c` — new window
`n` / `p` — next / previous window
`<number>` — go to that window
`w` — list windows
`%` — split left / right
`"` — split top / bottom
`arrows` — move between panes
`x` then `y` — kill the PANE
`&` then `y` — kill the WINDOW

10 Docker

`docker ps` — running containers
`docker ps -a` — all containers, stopped ones too
`docker images` — images on this machine
`docker logs -f <name>` — follow the output of a container
`docker exec -it <name> bash` — shell inside a running container
`docker start <name>` — start a stopped container
`docker stop <name>` — stop it
`docker rm -f <name>` — delete it
`docker run` takes an **image**, `docker start` takes an existing **container**.
Your container is called `duckiebot-yournick`.

11 When it does not work

`ros2: command not found`	`source /opt/ros/humble/setup.bash`
`Package 'blinker' not found`	not built or not sourced: `make build`, then `source install/local_setup.bash`
`can't open file '/opt/ros/humble/_local_setup_util_sh.py'`	you sourced `install/setup.bash` — use `install/local_setup.bash`
`VEHICLE_NAME is not set`	export it again: every new shell needs it
`ros2 topic list` shows nothing from the robot	use `make run`, not your own `docker run`
Edited a `.py` but nothing changed	restart the node
Nodes start, robot does not move	`ros2 topic echo /duckieXX/wheels_cmd` to see if commands arrive; check battery and emergency stop
LEDs flicker or ignore you	two nodes publish to `led_pattern`: stop `blinker` while driving